

the IoT infrastructure efficiently is also becoming more important. The operating systems in normal computers perform the primary functions such as resource management, user interaction, etc. The requirements of IoT operating systems are specialised due to the nature and size of the devices involved in the process. Some of the important characteristics/requirements of IoT operating systems are listed below (Figure 2):

- A tiny memory footprint
- Energy efficiency
- Connectivity features
- Hardware-agnostic operations
- Real-time processing requirements
- Security requirements
- Application development ecosystem

As of 2019, there is a spectrum of choices for selecting the operating system (OS) for the Internet of Things. Some of these OSs are shown in Figure 3.

Ubuntu Core

As Ubuntu is a popular Linux distribution, the Ubuntu Core IoT offering has also become popular. Ubuntu Core is a secure and lightweight OS for IoT, and is designed with a 'security first' philosophy. According to the official documentation, the entire system has been redesigned to focus on security from the first boot. There is a detailed white paper available on Ubuntu Core's security features. It can be accessed at https://assets.ubuntu.com/v1/66fcd858-ubuntu-core-security-whitepaper.pdf?_ga=2.74563154.1977628533.1565098475-2022264852.1565098475.

Ubuntu Core has been made tamper-resistant. As the applications may be from diverse sources, they are given privileges for only their own data. This has been done so that one poorly designed app does not make the entire system vulnerable. Ubuntu Core is 'built for business', which means that the developers can focus directly on the application at



Figure 2: IoT operating system features

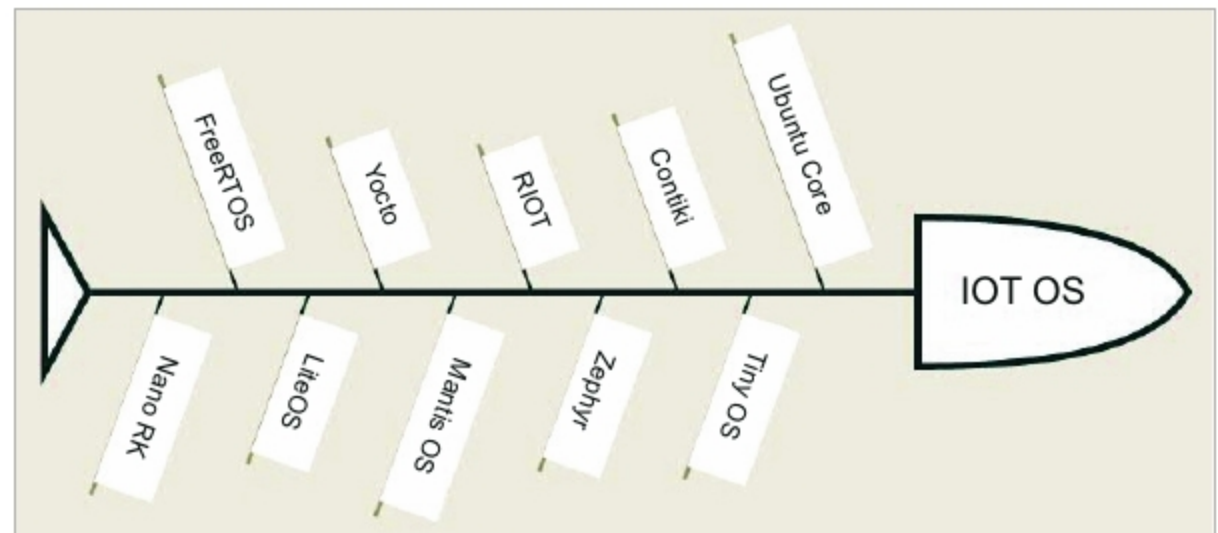


Figure 3: IoT operating systems

hand, while the other requirements are supported by the default operating system.

Another important feature of Ubuntu Core is the availability of a secure app store, which you can learn more about at <https://ubuntu.com/internet-of-things/appstore>. There is a ready-to-go software ecosystem that makes using Ubuntu Core simple.

The official documentation lists various successful case studies about how Ubuntu Core has been successfully used.

RIOT

RIOT is a user-friendly OS for the Internet of Things. This FOSS OS has been developed by a number of people from around the world.

RIOT supports many low-power IoT devices. It has support for various microcontroller architectures. The official documentation lists the following reasons for using the RIOT OS.

- **It is developer friendly:** It supports the standard environments and tools so that developers need not go through a steep learning curve. Standard programming languages

such as C or C++ are supported. The hardware dependent code is very minimal. Developers can code once and then run their code on 8-bit, 16-bit and 32-bit platforms.

- **RIOT is resource friendly:** One of the important features of RIOT is its ability to support lightweight devices. It enables maximum energy efficiency. It supports multi-threading with very little overhead for threading.
- **RIOT is IoT friendly:** The common system support provided by RIOT makes it a very important choice for IoT. It has support for CoAP, CBOR, high resolution and long-term timers.

Contiki

Contiki is an important OS for IoT. It facilitates connecting tiny, low-cost and low-energy devices to the Internet.

The prominent reasons for choosing the Contiki OS are as follows.

- **Internet standards:** The Contiki OS supports the IPv6 and IPv4 standards, in addition to the low-power 6lowpan, RPL and CoAP standards.